

Sustainable food vision for a low-carbon society



Liu Quanlin

Tianjin Tanggu NO.1 Middle School

CONTENTS

Chapter 1	Background Introduction.....	3
Chapter 2	Plan of Solutions	
2.1	Food Production.....	3
2.2	Food Processing.....	5
Chapter 3	Existed Problems.....	6
Chapter 4	Expected Effects&Epilogue.....	7
Chapter 5	Postscript.....	8

Chapter 1 Background Introduction

As an old saying goes, “People regard food as their heaven”. There’s no doubt that food plays an important role in our daily life. However, to meet the needs of the growing population, food production and processing practices have become more intensive and industrialized, which has caused a growing awareness of negative environmental impacts as a result. These negative impacts, including greenhouse gas emissions, water pollution, food security problems, etc., have gained attention from governments and experts from various fields.

In general, it’s vital to start with the perspective of food production and processing to conceive a sustainable food vision for a low-carbon society. This plan aims to explore strategies for achieving a sustainable food system that contributes to the goal of sustainable development.

Chapter 2 Plan of Solutions

2.1 Food Production

From the perspective of food production, here are some strategies for building a sustainable food vision for a low-carbon society.

2.1.1 Promote locally grown products. According to a research report published by the National Sustainable Agriculture Coalition, promoting local food systems has the potential to reduce transportation emissions and support local economies. In this part, governments and other related

institutions are expected to implement policies such as financial support programs helping to sustain local food production. Developing local planting can reduce the time and distance of long-distance transportation, lower carbon emissions, and increase the freshness and quality of food products.

2.1.2 Reduce food waste. According to CCTV News, about one-third of the world's food is consumed or wasted every year, with a total of approximately 1.3 billion tons per year. However, at least 820 million people out of the world's 7.366 billion population are facing hunger, equivalent to 1 out of every 9 people in the world going hungry, so food waste needs to be solved urgently. It is suggested that restaurants and consumers should follow the step of the Clean Plate Campaign to reduce food waste. Consumers are supposed to order food as much as they need and restaurants should encourage consumers to pick up the leftovers and take them away. With the amount of wasted food reduced, unnecessary pollution produced in the process of food production would decrease, which leads to a sustainable food vision.

2.1.3 Promote the use of green energy. It is suggested that farms and food factories should use renewable energy in order to reduce fossil fuel use and greenhouse gas emissions. What's more, promoting the use of green energy in food production is very promising and feasible. As the technology becomes more and more mature and its cost decreases, many

companies and governments agencies now encourage and support the use of green energy. In modern food production, the consumption of electricity, heat and fuel can be replaced by green energy such as solar, wind and hydrogen energy. At the same time, the use of green energy can reduce carbon emissions and environmental pollution. In a word, promoting the use of green energy is of great long-term significance for the healthy development of humans and nature.

2.2 Food Processing

From the perspective of food processing, here are some strategies for building a sustainable food vision for a low-carbon society.

2.2.1 Use sustainable packing materials. Sustainable packing materials can not only relieve environmental pollution but reduce resource consumption as well. Traditional packing materials like plastic bags can't break down, seriously affecting the permeability of the soil and hindering the growth of plants. Worse still, scientists predicted that before 2050, the amount of plastic bags would exceed the number of fish in the sea.

On the other hand, sustainable packing materials usually use new high-added-value materials that are recyclable and biodegradable, which contributes to environmental protection. What's more, using sustainable packing materials can enhance corporate image and market competitiveness. As sustainable lifestyle becomes more popular among consumers, food processing companies that use sustainable packing

materials will attract more attention and have better development prospects. At the same time, consumers are encouraged to reuse packing or use recyclable packing materials to reduce waste and carbon emissions.

2.2.2 Improve processing efficiency. By following this idea, energy consumption and waste will be reduced. With the development of technology and equipment, there are now many high-tech machines that can be used in food processing to improve processing efficiency. It is suggested that food processing companies should optimize production processes and equipment, use energy-saving equipment and more efficient processing techniques to reduce energy and water use, and lower carbon emissions. Automatic processing lines, robotic arms and computer-controlled systems can shorten the processing time and save energy. However, it should be mentioned that the improvement of processing efficiency cannot come at the expense of food equality and safety.

Chapter 3 Existed Problems

As we aim to create a sustainable food vision for a low-carbon society, there are still some problems existed in conducting the plan. Here are some of these:

3.1 Food waste. As mentioned earlier, food waste is a serious problem in

our daily life. As an old proverb says, “Old habits die hard”, there are still many reasons for consumers to waste food. They may order excessively to show face in front of others; they may be forced to buy food more than expected due to the unreasonable requirement of restaurants...All in all, there’s still a long way to go to provide food waste.

3.2 The high cost of green energy. Although using green energy in food production and processing have many benefits, the high cost of green energy could stop companies from its adoption. To promote the use of green energy in the food production and processing, governments and other related institutions are supposed to take actions such as financial and policy support for long-term development.

3.3 Limited access to sustainable materials. As an emerging industry, people still know little about sustainable materials. Sustainable packaging materials are not yet widely available or affordable, which makes it difficult for food processors to shift to sustainable packaging options. More investment and innovation are needed in the development and production of sustainable packaging materials to meet the demands of the market.

Chapter 4 Expected Effects&Epilogue

This plan starts from reality and builds a sustainable food vision for a low-carbon society. By conducting this plan, we can expect a number of

positive effects that contributes to building a low-carbon society, such as reduced gas emissions and environmental pollution, a decrease in the number of consumed resources, and raised awareness of people towards sustainable development. However, it should be mentioned that there are still some problems to overcome in implementing this plan. To address the challenges, institutions are supposed to take necessary actions, especially governments. With the conduction of this plan, we can expect a brighter future for food production and processing, where sustainability, quality and security are all achieved. It is the responsibility of every member of society to work together towards the vision for a sustainable future!

Chapter 5 Postscript

As a youth leader, I recognize that this plan has limitations due to time constraints and my limited skills. But anyway, I'm always on my way to learn and explore, to make contributions to society, no matter how small they may be.

I want to express my heartfelt gratitude to my guide teacher, Miss Han, for her continuous encouragement and assistance in helping me develop this plan. Her support has been invaluable in allowing me to showcase my ideas on this platform.

Thank you for taking your time to read this plan.